

Project Briefing for Mentors

How to explain the “ERT + GPR + Geotechnical + GIS” Dam Stability Project to your student — in plain language, with the conceptual framework, talking points, and milestones.

This document is a companion to the full technical guide (*Dam_Stability_GIS_Guide_Abidjan.pdf*). The technical guide tells the student which buttons to click. This briefing tells **you** how to make the project make sense to them.

Why You Need This Briefing

When a beginner GIS student is given a project that combines **geophysics**, **geotechnics**, and **spatial analysis**, the most common failure mode is not technical — it is conceptual. They click through tools without understanding why, produce plausible-looking maps, and cannot defend their choices in a viva.

This document gives you, the supervisor, the structured talking points to:

- bullet Frame the project as a single coherent story, not four disconnected techniques.
- bullet Pre-empt the conceptual confusions that beginners always hit.
- bullet Set clear milestones so progress is measurable, not just “working on GIS.”
- bullet Have ready answers to the questions students will ask (and the questions they should ask but won’t).

The 30-Second Elevator Pitch

Use this as the first thing you say to the student. Memorise it.

“Abidjan floods every rainy season. Some of those floods are made worse by failing earthen dams that nobody monitors. We want a map that says, for every dam in the district, *how likely is this one to fail, and who is downstream*. To get that, we look **inside** the dams using two geophysical techniques (ERT and GPR) and confirm with drilling (geotechnics). Then we put all of that, together with rainfall, slope, soil, and population, into one GIS model that ranks every dam from low to critical. The map is the deliverable. The decision-support is the impact.”

The Problem — Why This Project Exists

Context for Abidjan

- bullet Abidjan sits on a low-lying coastal sedimentary basin. Average annual rainfall is 1,400–2,400 mm, falling mostly in two intense rainy seasons.
- bullet The city has grown from ~1 million in 1980 to over 5 million today, with much of that growth in informal, unplanned neighbourhoods on flood-prone land.
- bullet Small earth dams (*barrages*) were built upstream of the lagoon system to manage run-off, irrigation, and water supply. Many are decades old, poorly documented, and never geotechnically re-assessed.
- bullet When a dam fails or overtops during a heavy rain event, the consequences are downstream flooding, loss of life, and infrastructure damage — on top of the flooding that would have happened anyway.

The research gap

Existing flood-risk studies for Abidjan (e.g., Kouamé et al. 2016) treat the surface only — rainfall, slope, drainage density, land use. None of them ask whether the dams themselves are part of the hazard. That is

the gap this project fills: **treating dam stability as a spatial, measurable variable** in the flood-risk equation.

The Four Pillars — Explained Simply

If your student remembers nothing else, they should remember what each pillar contributes and why we cannot remove any of them.

Pillar 1 — ERT (Electrical Resistivity Tomography)

Plain-language analogy: imagine taking the X-ray of a loaf of bread to see whether the inside is dry crumb or soggy. ERT does the same to the dam: it injects electrical current through metal stakes pushed into the ground and measures how easily the current flows.

- bullet **What it measures:** resistivity, in ohm-metres ($\Omega\cdot\text{m}$).
- bullet **What that tells us:** wet/clay material conducts well (low resistivity, $< 50 \Omega\cdot\text{m}$). Dry, granular, or rocky material resists current (high resistivity, $> 200 \Omega\cdot\text{m}$).
- bullet **For a dam:** a band of unexpectedly low resistivity inside the embankment is a red flag — it could be a saturated zone, internal erosion, or a seepage path.
- bullet **Limitation:** ERT is good at finding wet versus dry but cannot tell you *why* a zone is wet. That is what GPR and boreholes are for.

Pillar 2 — GPR (Ground Penetrating Radar)

Plain-language analogy: a fish-finder, but for the ground. GPR sends radar pulses into the soil and listens for echoes off any boundary where material properties change.

- bullet **What it measures:** reflected radar amplitude versus time (which we convert to depth).
- bullet **What that tells us:** the location and shape of interfaces — soil layering, voids, buried objects, the water table, cracks.
- bullet **For a dam:** bright, anomalous reflections indicate voids (very dangerous), buried structures, or sharp material changes that should not be there in a homogeneous embankment.
- bullet **Limitation:** GPR struggles in highly conductive (salty, clayey, wet) ground — exactly where ERT performs best. The two techniques are **complementary**, not redundant.

Pillar 3 — Geotechnical Investigation

Plain-language analogy: the ground-truth. ERT and GPR are both *indirect* — they infer subsurface properties from physical signals. Geotechnical drilling actually pulls out the soil and tests it.

- bullet **What it measures:** SPT N-values (a measure of soil stiffness), moisture content, permeability, lithology logged metre by metre.
- bullet **What that tells us:** the mechanical and hydraulic properties of the dam body, with absolute certainty at the borehole location.
- bullet **For the project:** boreholes **calibrate** the geophysics. If ERT shows a low-resistivity zone and a borehole nearby has high moisture and low SPT, you have high confidence that zone is real and weak.
- bullet **Limitation:** a borehole is one column in a 3D body. You can only afford a handful of them. That is why we need the geophysics to fill in the space between.

The teaching point: emphasise that geotechnics is “truth at a point” and geophysics is “estimate over space.” Together they give “truth over space.”

Pillar 4 — GIS (Geographic Information System)

Plain-language analogy: a stack of transparent maps you can put on top of one another, where every pixel knows its real-world coordinates.

- bullet **What it does:** stores, aligns, and combines spatial data — rasters (satellite imagery, DEMs), vectors (roads, buildings, dam points), and tables.
- bullet **Why we need it here:** the dam stability question is ultimately a *spatial* question. We are asking “where on the map is risk highest?” That is exactly what GIS is built to answer.
- bullet **For this project:** GIS is the integration layer. It is where ERT, GPR, geotechnical, rainfall, slope, soil, land use, and population all become *numbers in the same grid cells* that we can multiply, add, and rank.
- bullet **Limitation:** garbage in, garbage out. GIS will happily produce a beautiful map of nonsense if the inputs are wrong.

How the Four Pillars Combine — The Integration Logic

Walk through this with your student using a whiteboard. The diagram below describes the data flow:

Input layer	Source	Becomes
ERT inverted resistivity	Field survey	Surface resistivity raster
GPR depth slices	Field survey	Amplitude anomaly raster
Borehole SPT & moisture	Field drilling	Geotechnical weakness raster
DEM (SRTM)	Open data	Slope raster
DEM derivatives	Open data	Flow accumulation, drainage proximity
Rainfall (CHIRPS / WorldClim)	Open data	Mean annual rainfall raster
Land cover (BNETD / WorldCover)	Open data	LULC raster
Soil (HWSD / SoilGrids)	Open data	Soil texture / clay raster

All eight rasters above are reclassified to a common 1–5 scale (1 = low vulnerability, 5 = very high), then combined using AHP-derived weights into a single **Vulnerability Index** map. The geophysical and geotechnical layers carry the highest weights (~64% combined) because they describe the dam itself; the other layers provide environmental context.

The most important sentence to drill into your student: “The map is not the answer. The map is a hypothesis. Field validation is the answer.”

Key Concepts Your Student Must Understand

If you only have time to teach a few concepts, teach these. Each one is paired with a simple test you can use to check whether the student really gets it.

Concept 1 — Coordinate Reference Systems (CRS)

The idea: every spatial dataset describes locations using a coordinate system. Two systems that matter:

- bullet **Geographic CRS** (e.g., WGS84, EPSG:4326) — coordinates in degrees of latitude and longitude. Good for global storage. Bad for measurements: 1° of longitude is 111 km at the equator but 0 km at the poles.
- bullet **Projected CRS** (e.g., UTM Zone 30N for Abidjan, EPSG:32630) — coordinates in metres on a flat plane. **This is what you use for analysis.** Buffers, areas, and distances only make sense in projected CRS.

Test: ask the student “why are we using UTM 30N for Abidjan?” A correct answer mentions: (a) we need metres for buffers and areas, and (b) Abidjan’s longitude (~4°W) falls inside UTM Zone 30N’s coverage.

Concept 2 — Raster vs Vector

- bullet **Raster:** a grid of cells, each holding one value. Good for continuous phenomena (elevation, rainfall, resistivity). Example: SRTM DEM at 30 m.
- bullet **Vector:** discrete shapes — points, lines, polygons — with attribute tables. Good for discrete objects (a road, a dam, a commune boundary).

Most analysis in this project happens in **raster space**, because Weighted Overlay only works on aligned rasters. So every vector layer (e.g., drainage lines, dam buffers) eventually gets converted to a raster.

Test: “If I have a roads shapefile and want it in the vulnerability model, what do I do?” Correct: buffer → rasterise → reclassify.

Concept 3 — Reclassification (Why a 1–5 Scale?)

Each input layer measures something different in different units: resistivity in $\Omega \cdot m$, slope in degrees, rainfall in mm. We **cannot add** $47 \Omega \cdot m + 12 \text{ degrees} + 1850 \text{ mm}$. The numbers do not mean the same thing.

Reclassification solves this by mapping every layer onto a common dimensionless scale (we use 1–5, where 5 = highest vulnerability). After that, we can combine them.

Test: “Why do we reclassify ERT resistivity so that low values get high scores?” Correct: because low resistivity means saturated/weak zones, which is what we are *worried* about. The vulnerability scale is built around “higher number = worse.”

Concept 4 — AHP (Why Weights, Not Just Averages?)

If we simply averaged the eight reclassified layers, we would be saying that rainfall and ERT matter equally. They do not.

- bullet ERT and GPR describe the dam itself.
- bullet Rainfall and land use describe the surroundings.
- bullet Both matter, but they should not have the same weight.

AHP (Analytic Hierarchy Process) is a structured way to derive weights from **pairwise comparisons**. The student fills in a matrix asking, for each pair, “how much more important is X than Y?” The maths then gives back a consistent set of weights that sum to 1.

Critically, AHP also produces a **Consistency Ratio (CR)**. If $CR > 0.10$, the pairwise judgements contradict each other and the weights are unreliable. This is the safeguard against students just making up numbers.

Test: “What does CR > 0.10 mean and what do you do?” Correct: it means the pairwise judgements are inconsistent; revise the matrix until CR is below 0.10.

Concept 5 — Weighted Overlay

Once layers are reclassified (same 1–5 scale) and weights are set (sum to 100%), Weighted Overlay simply does:

$$\text{Vulnerability} = \sum (\text{layer}_i \times \text{weight}_i) \text{ for } i = 1..8$$

Every pixel ends up with one number between 1 and 5. We classify those into Low, Moderate, High, Very High, Critical and that is the final map.

Test: “A pixel scored 4.7. What does that mean and what should be done about that location?” Correct: very high vulnerability; recommend prioritised field inspection and monitoring.

Concept 6 — NoData (The Silent Map-Killer)

In raster analysis, **NoData** means “value missing here.” If even one input layer has NoData inside the study area, every output that uses it will also be NoData there. Beginners produce maps with mysterious blank patches and do not know why.

Common causes specific to this project:

- bullet Drainage proximity raster has NoData outside buffers → fix by reclassifying NoData → 1.
- bullet ERT/GPR/geotech surfaces only cover the immediate dam area → this is a real limitation; either extend the analysis extent to just the dam zones, or use a default value (e.g., 1) for “no information” areas with a clear caveat.
- bullet Soil rasters with sea pixels reaching into the lagoon → clip to the land boundary first.

Test: “Your final map has a hole in it. What are the three most likely reasons?”

Concept 7 — Validation (The Map Is a Hypothesis)

This is the concept students most often skip. A vulnerability map is a *prediction*. It must be tested.

Possible validation strategies for this project:

- bullet **Cross-validation between techniques:** do the ERT, GPR, and borehole anomalies agree at the locations where they overlap? If yes, confidence is high.
- bullet **Historical record:** do dams that historically had leakage, repairs, or near-failures fall in the High/Critical zones of the map?
- bullet **Expert review:** have a dam engineer or hydrogeologist look at the map and rate whether the rankings match their intuition.
- bullet **Sensitivity analysis:** change the AHP weights by $\pm 20\%$ and re-run. If the high-risk dams stay high-risk, the model is robust. If they shuffle wildly, the model is fragile.

Test: “Your map says Dam A is critical. How do you convince a sceptical engineer?” Correct answer involves at least two of the four validation strategies above.

Common Student Pitfalls (and How to Catch Them Early)

Each of the items below is a real failure mode that supervisors of beginner GIS projects see repeatedly. The “Catch it by” column tells you what to ask during a check-in to surface the problem before it ruins weeks of work.

#	Pitfall	Catch it by asking...
1	Working in geographic CRS by accident; buffers in ‘decimal degrees’ come out wrong by orders of magnitude.	“Show me the CRS of your active map and the unit on your scale bar.”
2	Mismatched cell sizes / extents between layers; Weighted Overlay fails or produces garbage.	“What is the cell size and snap raster you set in Environments?”
3	Reclassifying without justification — pulling break values out of thin air.	“Where do these breakpoints come from? Cite the literature or explain the physical reasoning.”
4	AHP matrix with consistency ratio > 0.10, ignored.	“What is your CR? Show me the calculation.”
5	Treating geophysical data as ground truth instead of an indirect estimate.	“What does an ERT low actually <i>mean</i> ? Could it be anything other than weakness?”
6	Producing a map and stopping. No validation, no sensitivity analysis, no caveats.	“How would you defend this map to a sceptical engineer?”
7	Cherry-picking only the data sources that confirm an expected pattern.	“Which dams disagree with your hypothesis, and what do you do about them?”
8	Beautiful cartography, weak analysis — spending three weeks on layout and ignoring methods.	“Walk me through your AHP weights before you touch the layout.”
9	Using a global dataset (HWSD soil, CHIRPS rainfall) at a spatial resolution that is meaningless for a single dam.	“What is the cell size of your input vs. the size of the object you are studying?”
10	Not backing up. Geodatabase corruption can wipe weeks.	“When did you last copy the .gdb folder offsite?”

FAQ — Likely Student Questions and Ready Answers

These are the questions you will be asked. Pre-loaded answers save mentoring time and give the student something concrete to write down.

Q1. “Why eight criteria? Why not five, or twelve?”

Because they are the ones that the literature on dam stability and on flood risk consistently identifies as primary. Three describe the dam itself (ERT, GPR, geotechnics), two describe the topography/hydrology context (slope, drainage proximity), and three describe the loading/environmental context (rainfall, LULC, soil). You can defend either direction — but if you add criteria, justify each one with a citation; if you remove, justify why it is captured by another.

Q2. “Where do my AHP pairwise scores come from?”

From three sources, in order of preference: (a) values published in peer-reviewed dam-stability or flood-risk AHP studies; (b) interviews with at least two domain experts (ideally an engineer and a geophysicist) recorded in your appendix; (c) your own reasoned judgement, clearly flagged. Whatever the source, the matrix must pass $CR < 0.10$.

Q3. “Can I do the project in QGIS instead of ArcGIS Pro?”

Yes, with caveats. QGIS has equivalents for almost every tool — SAGA and GRASS plugins cover hydrology, the *raster calculator* covers Weighted Overlay (you write the equation manually), and Python plugins handle AHP. The main loss is the polished cartographic layouts and Geostatistical Wizard. If your university does not have an ArcGIS licence, QGIS is fine; document the equivalent tools in a methods appendix.

Q4. “What if I cannot do field ERT/GPR/drilling?”

Then the project becomes a **methodology paper** rather than a case study. Run the GIS workflow with the open data (rainfall, slope, LULC, soil, drainage) and use synthetic or literature-derived values for the geophysical and geotechnical layers, clearly labelled as illustrative. The deliverable becomes “a reproducible workflow that a future field campaign can plug measurements into.” State this scoping clearly in the introduction.

Q5. “How accurate is my final map?”

Honestly, you do not know without validation. The map encodes your **assumptions** (which criteria, which weights, which breakpoints). Run a sensitivity analysis: change each weight by $\pm 20\%$ and see if the high-risk dams remain high-risk. If yes, the ranking is robust to the assumptions; if no, report which assumption drives the result. That nuance is what gets you the higher mark.

Q6. “The flood inundation in Phase 5 looks too simple. Is it real?”

It is a **bathtub model**: any pixel below the dam crest elevation, downstream, is flagged as potentially flooded. It ignores hydraulic dynamics, channel roughness, attenuation, and travel time. Honest students label it “maximum potential extent” rather than “flood prediction.” If a real inundation forecast is needed, that is a HEC-RAS project, not a GIS project.

Q7. “Can I use this work to actually advise the city?”

Not directly. A student-level vulnerability index is a **screening tool**: it flags candidate dams that warrant professional engineering inspection. The chain to actual policy advice runs through licensed dam-safety engineers and regulatory bodies. Frame the output that way and you are ethically and professionally clean.

The Workflow at a Glance (Use This for Your Whiteboard)

Six phases, one map. If your student can sketch this from memory, they understand the project.

Phase	What happens	Output
1. Data acquisition	Download open data; collect ERT, GPR, borehole field data.	Folder of source files
2. ArcGIS Pro setup	Project, environments, clip to study area, derive slope and hydrology.	Cleaned base layers + study area map
3. Geophysics integration	Georeference ERT/GPR, interpolate to surfaces, score boreholes.	Three subsurface rasters
4. Multi-criteria analysis	Reclassify all 8 layers; AHP weights; Weighted Overlay.	Vulnerability Index map
5. Decision-support outputs	Per-dam vulnerability, simplified flood inundation, population and infrastructure at risk.	Final thematic maps
6. Validation & report	Sensitivity analysis, expert review, write-up, defence.	Thesis / report

Milestones — What “On Track” Looks Like

Use these as gates. Do not let the student progress to the next milestone until the previous one is genuinely complete. Every milestone has a single artefact that proves it.

#	Milestone	Artefact that proves it	Approx. timeline
M 1	Project framing complete	One-page project brief with goals, study area, criteria list, data sources	End of week 2
M 2	Open data acquired and inspected	Map of Abidjan with all open layers loaded, projected, clipped	End of week 4
M 3	Field data collected and processed	ERT inverted models, GPR depth slices, borehole logs in CSV	End of week 7
M 4	All inputs integrated into GIS	Eight reclassified rasters in the geodatabase, all aligned	End of week 10
M 5	AHP matrix and weights validated	Pairwise matrix with CR < 0.10, weights table, sensitivity table	End of week 11
M 6	Final vulnerability map produced	Vulnerability_Index raster + per-dam summary table	End of week 13
M 7	Validation and decision-support	Sensitivity analysis, simplified inundation, exposure analysis	End of week 14
M 8	Report draft submitted	Full draft with figures, tables, references	End of week 16

If the student is two weeks behind on a milestone, that is the moment to intervene — not at week 16.

What ‘Good’ Looks Like — Final Deliverables Checklist

By the end of the project, the student should hand you **all** of the following. Treat the absence of any item as a fail.

Maps (export each as 300 DPI PDF and PNG)

- bullet Study area / location map (with inset showing Côte d’Ivoire)
- bullet Geology / soil map of Abidjan
- bullet Slope and rainfall maps
- bullet Land-use / land-cover map
- bullet Drainage and proximity buffer map
- bullet ERT cross-section per dam, plus interpolated surface map
- bullet GPR depth-slice map for each survey
- bullet Eight reclassified factor maps (1–5 scale)
- bullet Final vulnerability map (the headline figure)
- bullet Per-dam vulnerability ranking map
- bullet Simplified flood inundation map per critical dam
- bullet Exposure map (population & infrastructure at risk)

Tables

- bullet Data inventory table (source, format, resolution, date)
- bullet AHP pairwise comparison matrix and CR calculation
- bullet Final weights table with justification per factor
- bullet Sensitivity analysis table (weights $\pm 20\%$)
- bullet Per-dam vulnerability summary (mean, max, class)
- bullet Exposure summary (population, buildings, road km per dam)

Document & data

- bullet Full written report or thesis
- bullet Geodatabase (.gdb) with all final layers, named consistently
- bullet Project metadata: README listing every layer with source, processing steps, units, CRS, and date
- bullet Code or model: any Python scripts, ModelBuilder models, or AHP spreadsheets used
- bullet Field logs: photos, GPS tracks, raw geophysical data

Defensible narrative (the most important deliverable)

The student should be able to deliver a 10-minute presentation that answers, in order:

- 1 What problem does this solve and who cares? (1 minute)
- 2 What did you measure and where? (2 minutes)
- 3 How did you combine it? (2 minutes — AHP + Weighted Overlay)
- 4 What does the map say? (2 minutes — specific dam rankings)
- 5 How confident are you and why? (2 minutes — validation, sensitivity, limitations)

Suggested Mentoring Cadence

- bullet **Weekly 30-minute check-in:** progress against the current milestone, blockers, what is being done in the coming week.
- bullet **Bi-weekly 60-minute working session:** open ArcGIS Pro together, look at actual files, point to specific issues. This is where most learning happens.
- bullet **Monthly milestone review:** formal sign-off on the milestone artefact. Do not sign off if it is incomplete; rolling debt kills these projects.
- bullet **Mock viva at week 14:** 20 minutes of presentation plus 10 minutes of hostile questions. Surfaces every weakness while there is still time to fix it.

Closing Notes for the Mentor

- bullet **Resist the urge to drive the mouse.** When the student is stuck, narrate what they should do; do not take the keyboard. They learn by clicking, not by watching.
- bullet **Frame every check-in around the question, not the tool.** “What are you trying to find out?” beats “Show me the Reclassify dialog.”
- bullet **Reward the caveat.** Students who say “here is the map and here is what is wrong with it” should be graded above students who present an unqualified map, even if the latter looks prettier.
- bullet **Connect to practice.** A field visit to one of the barrages, or a guest seminar from a working dam engineer, is worth more than three lectures.

Companion document: *Dam_Stability_GIS_Guide_Abidjan.pdf* — the full step-by-step technical workflow your student should follow. This briefing is the ‘why’ behind that ‘how’.